Predicting Life Expectancy from Health, Immunization and Socioeconomic Factors

Using Linear Regression

AI101 Mini-Project | IIT Ropar

Team 29

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Repo
github.com/aaditverma0/life-expectancy-predictor

Problem and Data

What and why

- Predict life expectancy (in years) from 20 indicators; 17 kept after cleaning
- Shows which factors matter most, to guide health policy

Dataset source

WHO Global Health Observatory data, via Kaggle: "Life Expectancy (WHO)"

2,938
rows

22
columns

193
countries

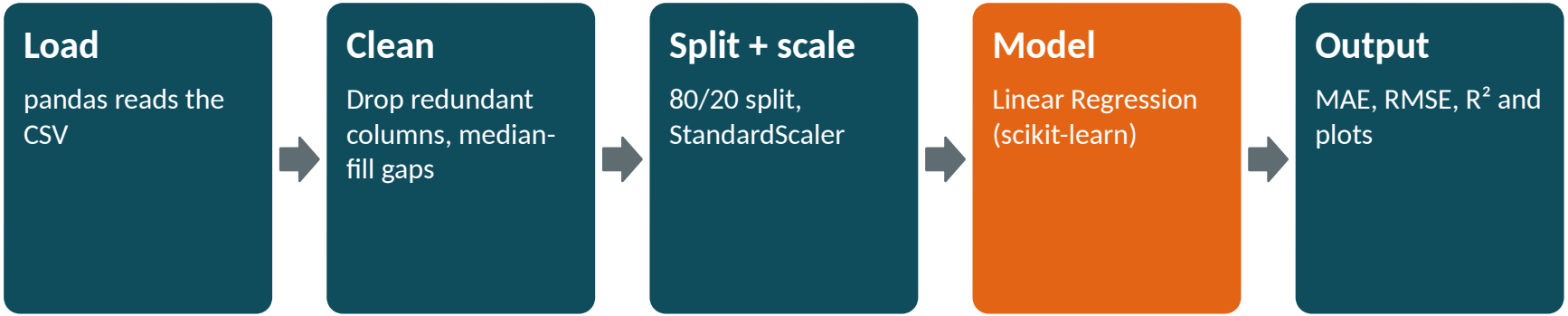
2000-2015
years covered

Health
Adult mortality, BMI, HIV/AIDS, measles, thinness

Immunization
Hepatitis B, Polio, Diphtheria

Socioeconomic
GDP, schooling, income composition, status

Approach



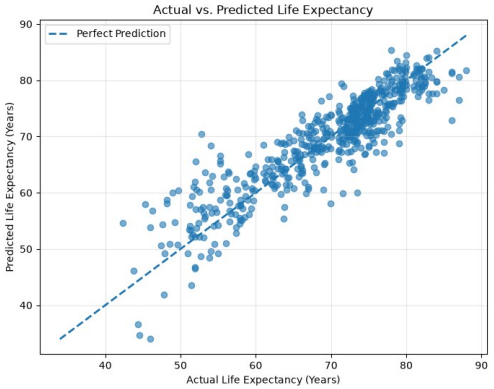
Libraries used

pandas | NumPy | scikit-learn | Matplotlib | Seaborn

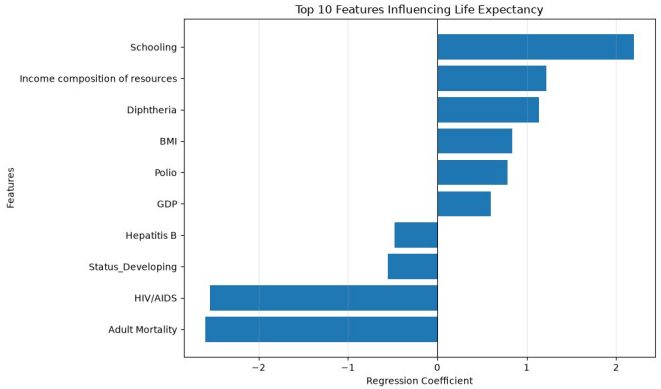
Input: data CSV → Output: metrics + plots

Results

Predictions track actual values closely



Adult Mortality, HIV/AIDS and Schooling matter most




```
Model Evaluation Results
MAE          : 3.0163
RMSE         : 4.0583
R² Score     : 0.8096
5-fold CV R²: 0.8066 ± 0.0121
```

R^2	RMSE	MAE
0.81	4.06 yrs	3.02 yrs

Learnings

1 What worked, what didn't

- Linear regression: simple, interpretable baseline
- Correlated features (e.g. infant vs under-five deaths) had to be dropped
- Non-linear effects limit accuracy (R^2 0.81)

2 Who did what

- **Aadit**: data exploration and cleaning
- **Garima**: data split and model building
- **Mishika**: validation and evaluation
- **Jayant**: result plots, demo, final run

3 AI-use disclosure

- Claude: starter script and slide draft
- We ran, tested and can explain all code
- No AI generated the data or results